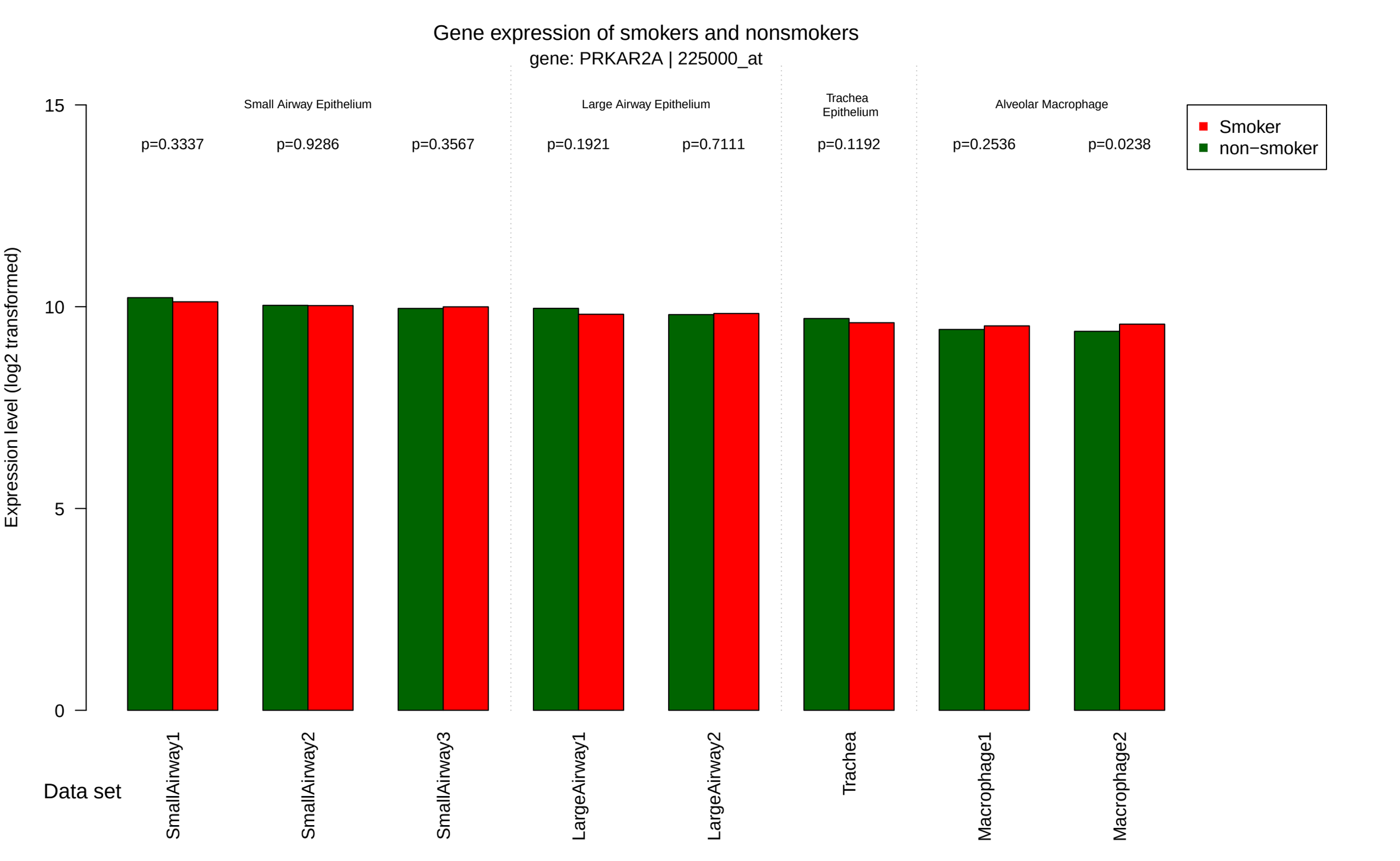
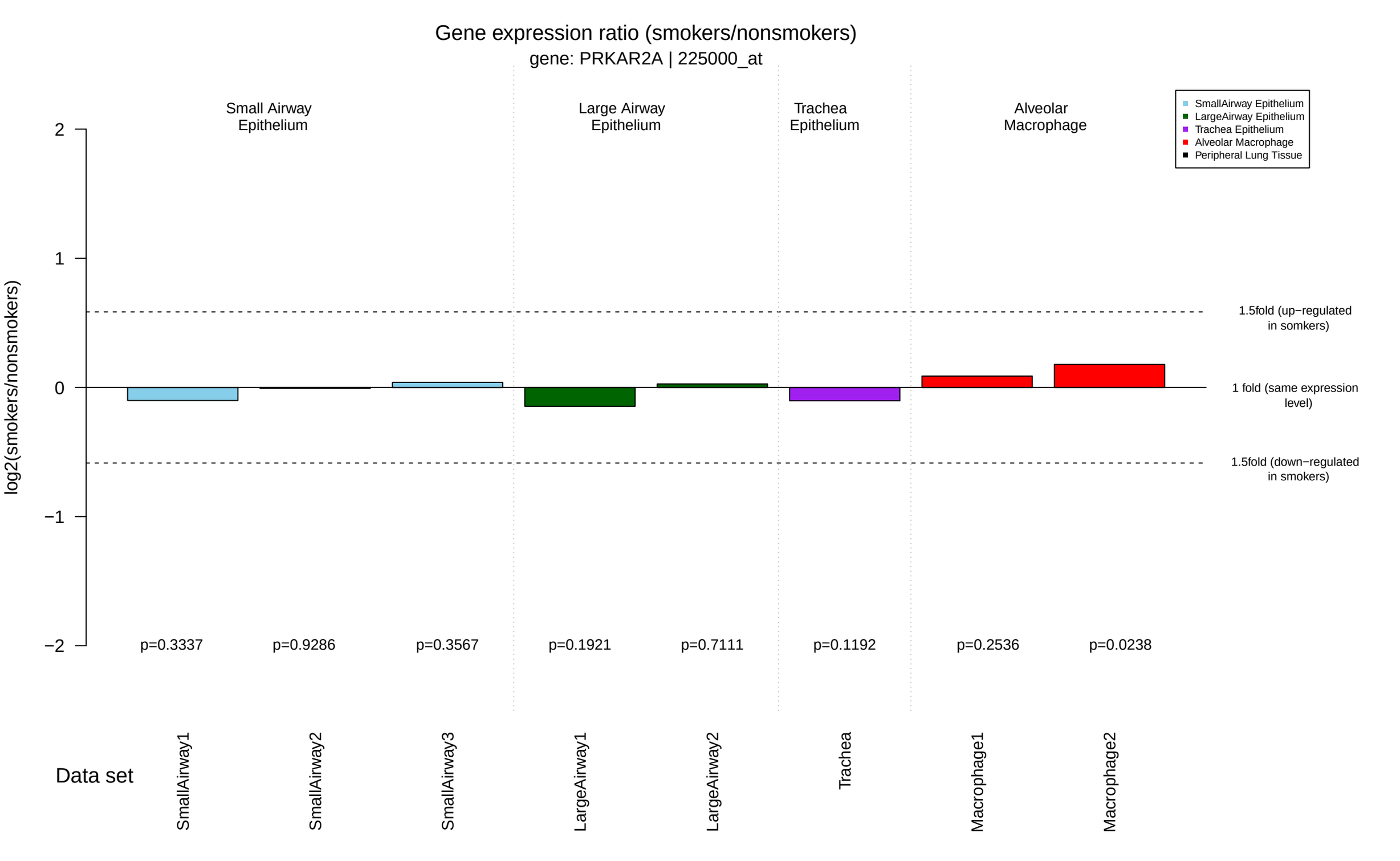
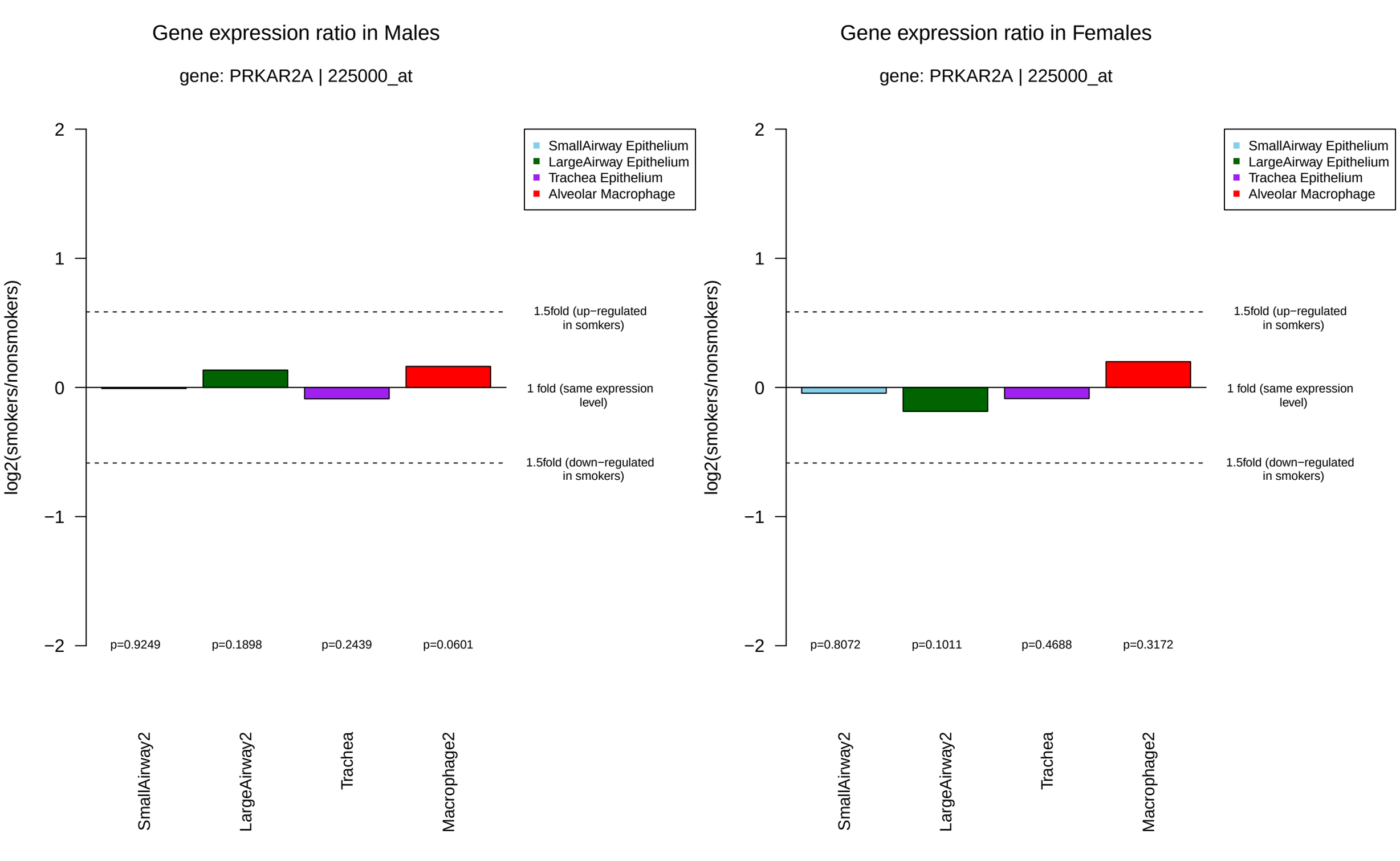
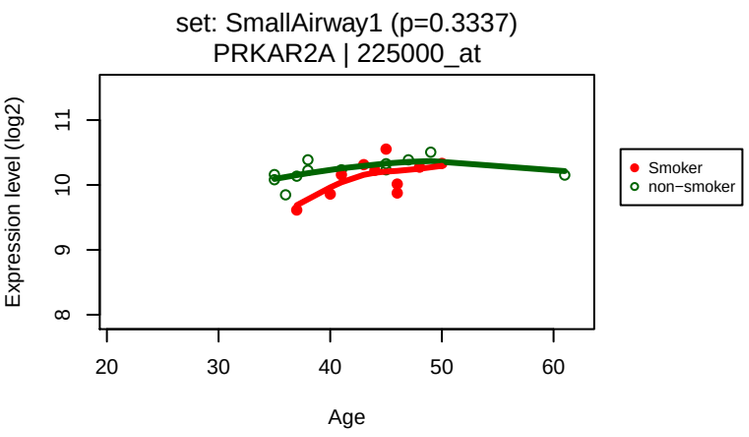
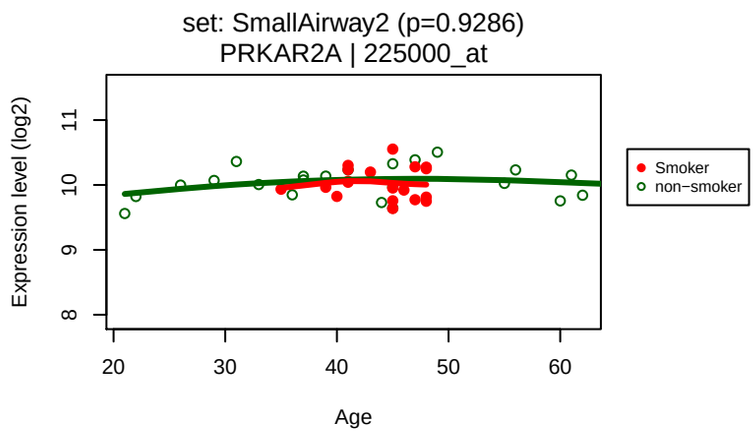
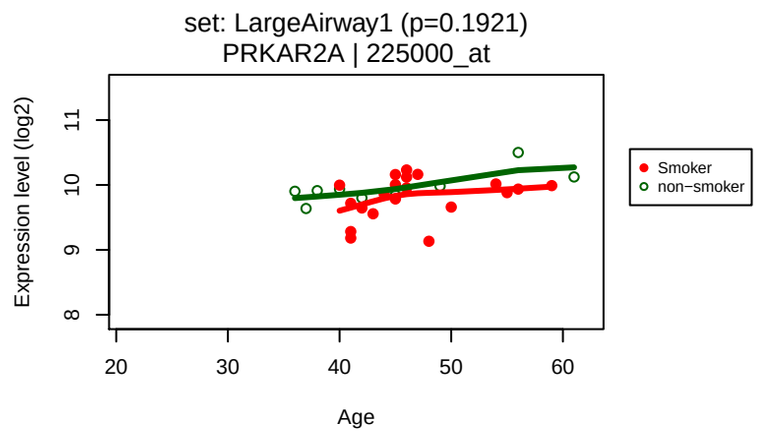

Gene expression of smokers and nonsmokers
gene: PRKAR2A | 225000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3337
p=0.9286
p=0.3567
p=0.1921
p=0.7111
p=0.1192
p=0.2536
p=0.0238
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

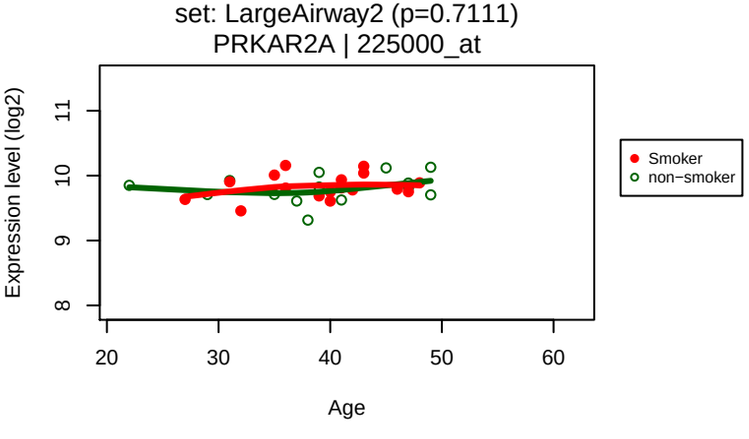

Gene expression ratio (smokers/nonsmokers)
gene: PRKAR2A | 225000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3337
p=0.9286
p=0.3567
p=0.1921
p=0.7111
p=0.1192
p=0.2536
p=0.0238
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

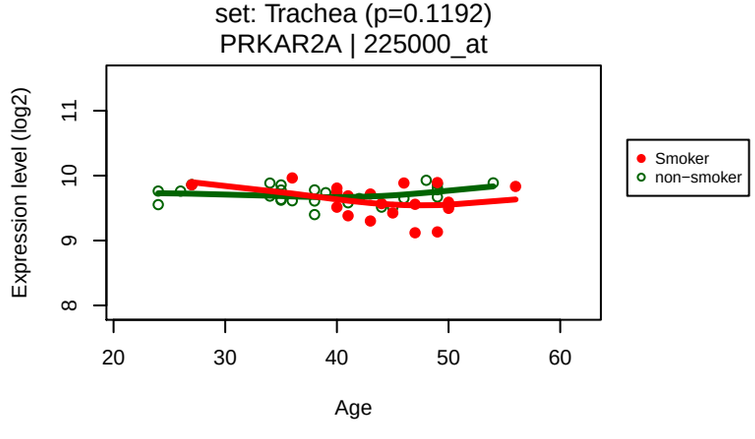

Gene expression ratio in Males
gene: PRKAR2A | 225000_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9249
p=0.1898
p=0.2439
p=0.0601
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAR2A | 225000_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8072
p=0.1011
p=0.4688
p=0.3172
SmallAirway2
LargeAirway2
Trachea
Macrophage2

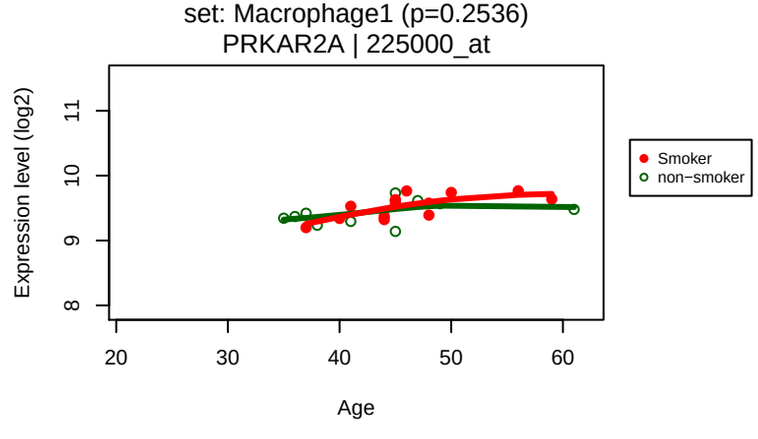

set: SmallAirway1 (p=0.3337)
PRKAR2A | 225000_at
Expression level (log2)
Age
Smoker
non−smoker

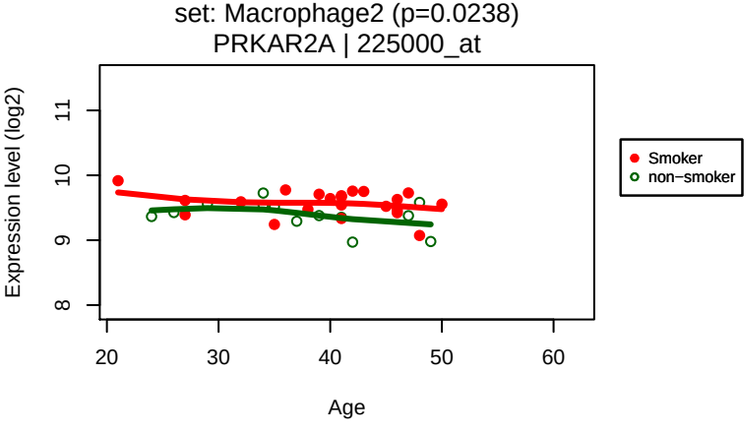

set: SmallAirway2 (p=0.9286)
PRKAR2A | 225000_at
Expression level (log2)
Age
Smoker
non−smoker

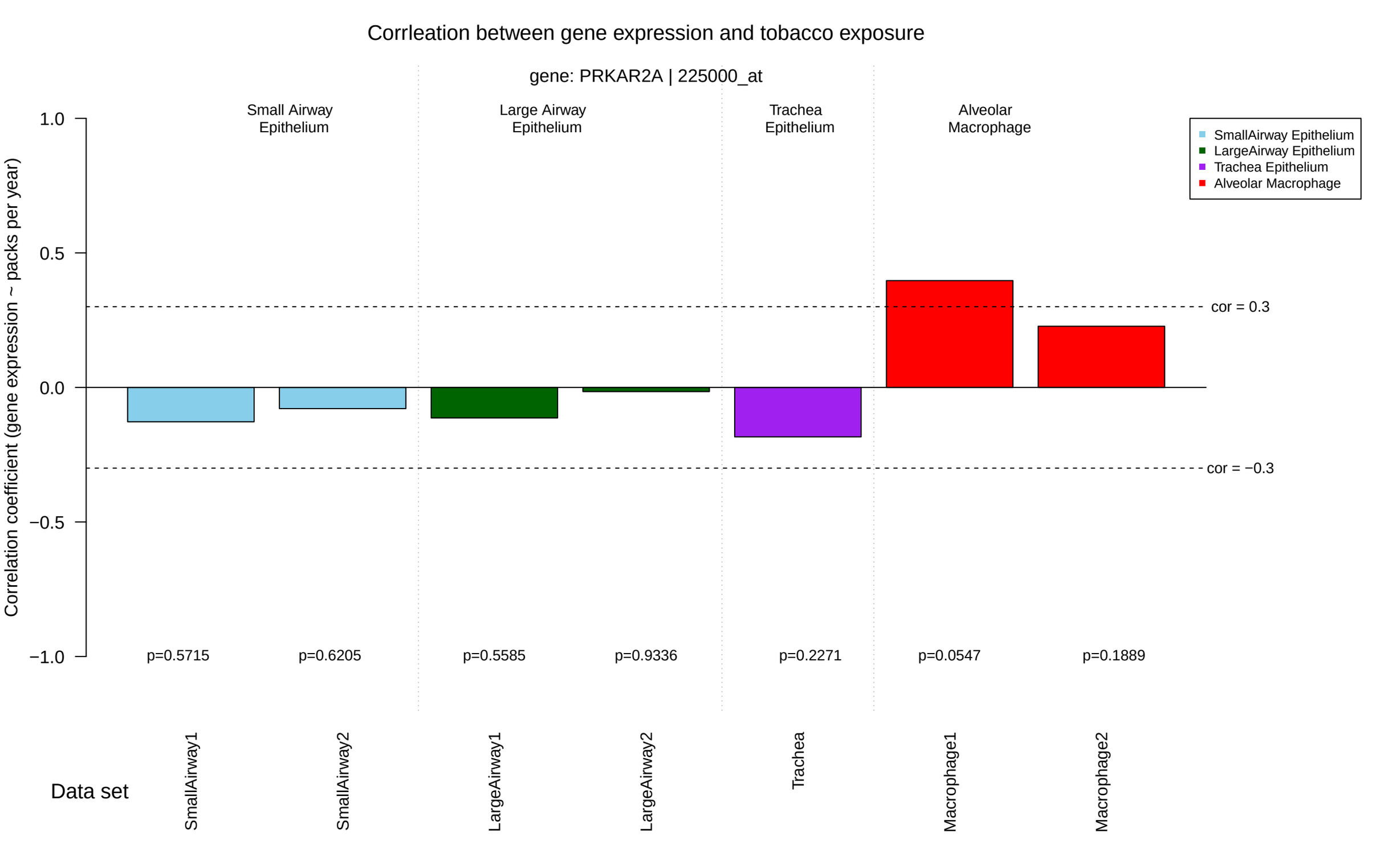

set: LargeAirway1 (p=0.1921)
PRKAR2A | 225000_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7111)
PRKAR2A | 225000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1192)
PRKAR2A | 225000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2536)
PRKAR2A | 225000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0238)
PRKAR2A | 225000_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAR2A | 225000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5715
p=0.6205
p=0.5585
p=0.9336
p=0.2271
p=0.0547
p=0.1889
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2